Memory Makes the Poison: Over Memorization Drives Visual Data Poisoning in LVLMs

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Abstract

Large Vision–Language Models (LVLMs) excel across tasks, yet their safety and security remain underexplored. Among emerging threats, LVLMs poisoning attacks pose a serious threat by inducing targeted hallucinations in fine-tuned LVLMs. Although effective, the root cause of these attacks remains poorly understood. The attack is originally justified as being effective due to the carefully injected visual perturbations to fine-tuning data, which subtly manipulate the model. Consequently, existing defenses rely on state-of-the-art (SOTA) purification methods, but these have shown ineffective so far. In this work, we argue that this gap stems from a more fundamental issue: a limited understanding of LVLM vulnerabilities during fine-tuning. To address this, we systematically study the fine-tuning process and, for the first time, identify over-memorization as the key vulnerability: *LVLMs tend to over-memorize fine-tuning concepts, directly leading to hallucinations in fine-tuned models. Our finding overturns the original justification: the dominant driver is* ***over-memorization*** *of injected concepts, not the visual perturbation.* Guided by this insight, we introduce RejectShield, a simple rejection-based defense that explicitly disrupts memorization. Across eight settings spanning attack variants, attack goals, model families, and access regimes, RejectShield reduces attack success by up to 99% while largely preserving normal performance. Finally, we discuss broader implications of this memorization vulnerability, including evaluation methods that test concept replay and training practices that mitigate memorization pressure.

Keywords

Data Memorization, Data Poisoning, Large Vision Language Model, Adversarial Examples

1 Introduction

Research Gap. Large Vision–Language Models (LVLMs) are routinely fine tuned on mixed, crowd sourced data, which opens a training time attack surface: *visual perturbation based data poisoning* [2, 12, 13, 20]. For example, in SOTA attack ShadowCast [20], an adversary adds edits (i.e., imperceptible perturbation) to a small fraction of training images so that, after deployment, the model produces targeted hallucinations on clean inputs. Original works attribute their success to the strength of the perturbations themselves, leading defenses to concentrate on purifying poisoned samples. Yet, existing defenses remain ineffective, particularly against SOTA attacks on LVLMs [20]. In this paper, we argue that this gap stems from a more fundamental issue: a limited understanding of the root causes of LVLM vulnerabilities.

Key finding: We discover that the dominant driver of poisoning is *over-memorization* of concept cues during fine tuning, not the visual perturbation. Through carefully designed experiments, we show that LVLMs are prone to over-memorizing fine-tuning concepts, which directly leads to hallucinations in fine-tuned models. This perspective reframes visual perturbation poisoning: its effectiveness arises not merely from imperceptible perturbations, but fundamentally from the excessive memorization tendencies of LVLMs. We further observe that multimodality amplifies this effect relative to unimodal counterparts.

From insight to defense. If poisoning sticks because the model over memorizes a small, concept aligned subset of samples, defenses should reduce memorization pressure rather than only clean pixels. We introduce **RejectShield**, a simple yet effective rejection-based defense that mitigates poisoning by disrupting memorization of corrupted concepts. Across extensive experiments, our method significantly reduces attack success rates while preserving model utility, consistently outperforming existing defenses. Beyond poisoning, our findings highlight data memorization as a general, fundamental vulnerability of LVLMs, exposing new attack surfaces and pointing toward pathways for principled defenses. In summary, our main contributions are as follows:

- **Reveal concept over-memorization as a novel LVLM vulnerability during fine-tuning.** We show that LVLMs tend to over-memorize fine-tuning concepts, leading to hallucinations. Moreover, this effect is amplified in multimodal settings, highlighting a previously unexplored dimension of LVLM vulnerability. (See Sec. 3)
- **Comprehend understand of SOTA data poisoning attacks on LVLMs and propose RejectShield as an effective defense.** Contrary to the original justification, our findings reveal the underlying mechanisms that enable recent poisoning attacks on LVLMs [20] and explain why subsequent existing purification-based defenses (built upon that justification) ultimately fail. To address this gap, we propose RejectShield, a simple yet effective defense that protects training data during LVLM fine-tuning. Extensive experiments spanning four attack goals, three LVLM architectures, diverse prompts, attack variants, and both black-box and white-box settings demonstrate that RejectShield consistently outperforms prior defenses, reducing attack success rates by up to 99% while largely preserving model utility (see Sec. 4.1)
- **Highlight broader safety implications.** Beyond poisoning, we discuss the wider impact of the newly identified data memorization vulnerability on LVLM data poisoning, expose

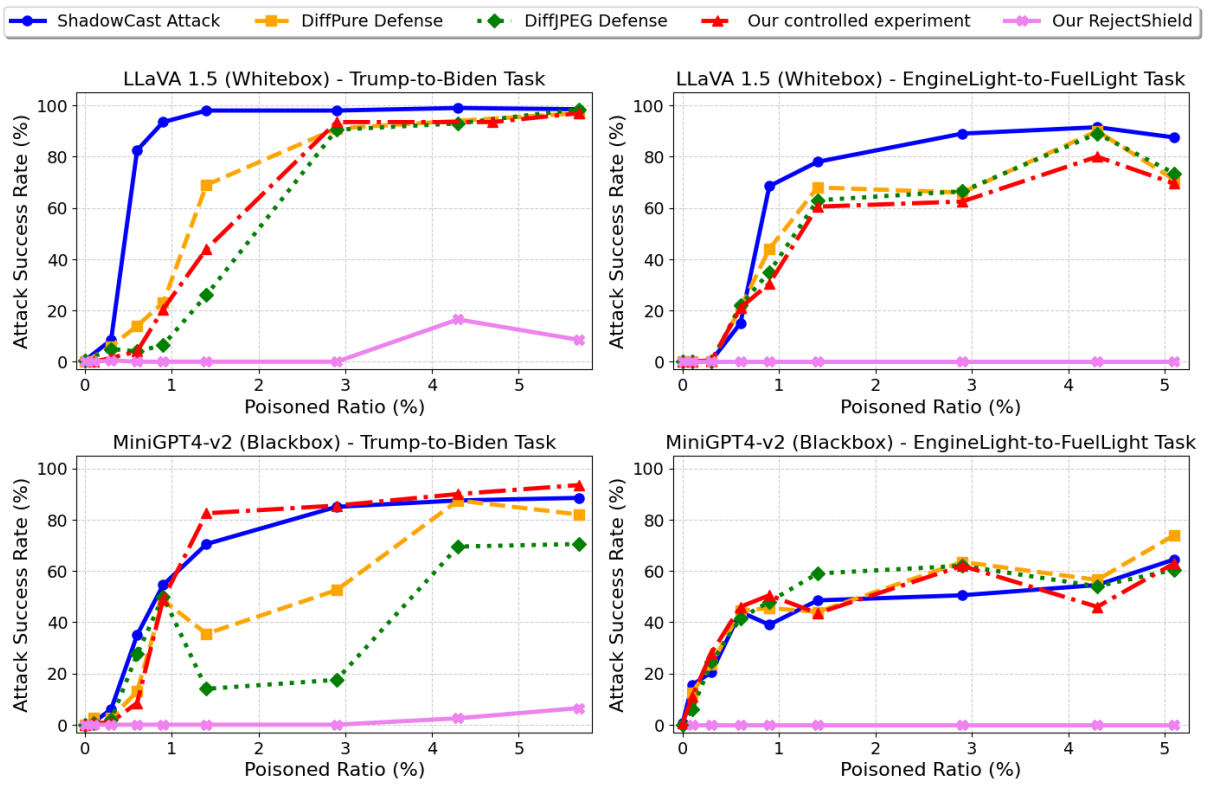

Figure 1: (1) We identify concept over-memorization as a novel LVLM vulnerability during fine-tuning. Controlled experiments (red) isolate the effect of data memorization during fine-tuning, showing that LVLMs over-memorize injected concepts, leading to hallucinations in fine-tuned LVLMs. Detailed setups and results are provided in Sec. 3 and the Appendix. (2) Our discoveries comprehend understanding of visual perturbation-based data poisoning on LVLM and explain why existing purification-based defenses fail. Visual perturbation-based data poisoning attacks including such SOTA attack on LVLM [20] originally attributed to visual perturbations, motivating existing defenses to focus on purification. However, our discoveries show memorization is a critical hidden factor driving attack success. This explain why current purification-based defenses are ineffective. (3) Our proposed RejectShield. Leveraging our insights, we propose rejection-based RejectShield (pink). Our extensive results show that RejectShield effectively reduces attack success rates by up to 99% while nearly preserving model utility. Detailed results and experimental setups are provided in Sec. 4.1

new potential attack surfaces, and outline pathways toward more robust defenses. (See Sec. 4.2)

2 Related Work

Data Poisoning Attack in LVLMs. Data poisoning attacks aim to inject malicious data during training to compromise a model's behavior at inference [3, 15, 24]. Prior research has primarily focused on a unimodal setting, such as vision-only [13, 17, 23] or text-only models [7, 18]. In the multimodal setting, a recent pioneering work, ShadowCast [20], demonstrates that the fine-tuning stage, while crucial for adapting LVLMs to downstream tasks, can be highly vulnerable to data poisoning attack. ShadowCast introduces an attack that injects imperceptible visual perturbations into LVLMs fine-tuning data, leading to targeted hallucinations in fine-tuned LVLMs. Despite its demonstrated effectiveness, the underlying reason for the attack's success remains poorly understood. As a result, existing defenses that focus only on purifying visual perturbations are ineffective in defending against ShadowCast attack [20]. In this work, we present the first in-depth investigation of ShadowCast and uncover a critical but previously overlooked cause of this attack's success: data memorization during LVLMs fine-tuning. Building on this insight, we propose a simple yet effective rejection-based defense that significantly mitigates the attack while nearly maintaining model utilities.

Memorization in Deep Neural Networks. While commonly believed to learn general patterns for specific tasks, DNNs can also memorize training data, even with random labels [21]. This suggests that DNNs may memorize specific features of individual examples rather than extracting generalizable patterns. In practice, such behavior risks encoding these artifacts, ultimately compromising the model's security. For LLMs, previous works have demonstrated that models can memorize and even leak rare or sensitive information from their training corpora [1, 8, 22]. Extending these concerns to multimodal settings, [6] showed that contrastive vision-language

models like CLIP can recall fine-grained visual details from training images, even when such details are absent from the captions. However, their analysis primarily focuses on image-text retrieval tasks and does not address the behavior of generative LVLMs, which produce free-form outputs. In this work, we present a pioneering investigation into data memorization in generative LVLMs. Our findings reveal a distinct and underexplored failure mode: the tendency of LVLMs to hallucinate answers by relying on memorized fine-tuning data patterns rather than grounding their responses in the actual multimodal input (e.g., an image and a corresponding question) at inference time. This phenomenon highlights a critical gap in our understanding of LVLM behavior and underscores the need for future work focused on improving grounding and factual consistency in generative outputs in the fine-tuning stage.

3 Over-Memorization during LVLM Fine-tuning

LVLMs are shown to be vulnerable to data poisoning attacks [20]. However, the underlying reason for the success of such an attack is poorly understood due to the limited understanding of LVLM vulnerabilities during fine-tuning. In this section, we investigate LVLM vulnerability during fine-tuning through the lens of data memorization. First, we outline the motivation for our study in Sec. 3.1. Then, in Sec. 3.2, we design controlled experiments that isolate the role of data memorization in fine-tuning, showing that LVLM tend to over memorize injected concepts, which leads to persistent hallucinations, especially as the number of injected samples increases. In Sec. 3.3, we extend this analysis to compare unimodal and multimodal settings, demonstrating that multimodal inputs exacerbate data memorization. In summary, our findings uncover a previously unexplored dimension of LVLM vulnerability.

3.1 Background and Motivation

Fine-tuning is a critical step for enhancing LVLM performance on downstream tasks. However, during this process, LVLMs remain vulnerable to malicious or uncurated data. Ensuring robustness in fine-tuning is particularly crucial when fine-tuned LVLMs are deployed in high-stakes applications. Yet, our understanding of vulnerabilities in LVLM fine-tuning remains limited. This crucial gap limits our understanding of root causes of certain attacks. Concretely, the SOTA data poisoning attack, ShadowCast [20], poses a severe threat to LVLM by inducing targeted and stealthy hallucinations. Despite requiring only a few poisoned samples, ShadowCast achieves high attack success (see Fig. 1), and *critically, no effective defenses currently exist.*

Particularly, ShadowCast defines its malicious objective using a destination concept C_d (e.g., "Joe Biden") and an original concept C_o (e.g., "Donald Trump"), with the goal of making the fine-tuned LVLM misidentify C_o as C_d. Starting from a benign example (x_d, y_d) of C_d, a poisoned image x_p is optimized as:

$$x_p = x_d + \delta^*, \tag{1}$$
$$\delta^* = \arg\min_{\|\delta\|_\infty \le \epsilon} \|\phi(x_d + \delta) - \phi(x_o)\|_2^2,$$

where ϕ is the vision encoder in the LVLM, mapping images into the shared multimodal embedding space. The optimization ensures that x_p (i) remains visually indistinguishable from x_d, while (ii) in the LVLM's feature space, it aligns with C_o. Finally, x_p is paired with y_d to form the poisoned example (x_p, y_d). Training on such samples causes the LVLM to associate features of the original concept image x_o with the target label y_d, resulting in targeted hallucinations at inference (e.g., mislabeling Trump as Biden). These few poisoned examples, $\mathcal{D}_{poison}$: (x_p, y_d), are then injected to the clean data, $\mathcal{D}_{clean}$: (x_c, y_c), to form the final training dataset, $\mathcal{D}_{train}$, used for LVLM fine-tuning.

Inspired by a unimodal study [13], ShadowCast is originally justified its success solely by the effect of visual perturbations. Consequently, existing defenses primarily focus on sanitizing visual perturbation [10, 14]. However, these methods fail to effectively reduce attack success rates (see [20] and Fig. 1). This raises a fundamental question: *even after addressing the core mechanism of SOTA attacks, why do defenses still fail? Could there be an underexplored factor at play?*. We argue this reflects a deeper issue, compared to unimodal models, the fundamental vulnerabilities of multimodal LVLM remain poorly understood. This lack of understanding risks mischaracterizing the nature of LVLM attacks and misguiding the development of effective defenses.

3.2 Concept Over-Memorization as a previously-unknown yet critical vulnerability during LVLM fine-tuning

Building on the motivation in Sec. 3.1, we hypothesize that LVLMs tend to over-memorize injected concepts during fine-tuning, even when only **a few benign examples** are present. To validate this, we now take a step back to examine LVLM vulnerabilities through the lens of data memorization. Particularly, we design a systematically controlled experiment that isolates the role of data memorization during LVLM fine-tuning. Our investigation reveals, for the first time, that data memorization is a previously-unknown yet critical vulnerability in LVLM during fine-tuning. This finding not only deepens our understanding of LVLM risks but also uncovers a root cause behind data poisoning attacks, offering new directions for effective defenses. More broadly, we show that data memorization is a general vulnerability of LVLM, highlighting an urgent need for further research in this area.

Experimental Design. In our controlled experiments, we adapt the experiment in [20] including model architectures (LLaVA 1.5 and MiniGPT-v2), fine-tuning hyperparameters, training procedure, the cc-sbu-align dataset [25] as the downstream dataset, and the evaluation protocol. The only modification is the injected samples during fine-tuning. Here, instead of adversarially perturbed images x_p, we fine-tune the LVLMs with their benign counterparts x_d, while keeping all other inputs identical. This design isolates the effect of data memorization, ensuring that any vulnerability observed is solely due to memorization during fine-tuning. We evaluate on the Trump-to-Biden and EngineLight-to-FuelLight tasks, using the same poisoned-sample ratios and training schedules as in [20].

The detailed results are shown in Fig. 1. First, we observe that for both LLaVA 1.5 and MiniGPT-v2, across the *Trump-to-Biden* and *EngineLight-to-FuelLight* tasks, the attack success rate of our controlled experiment (red) achieves very high success rates. Notably, even at a low injection ratio of 1%, MiniGPT-v2 reaches an attack

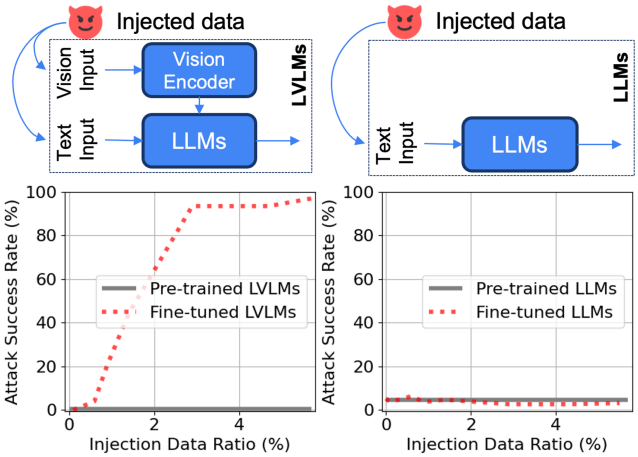

Figure 2: Our investigation on the data memorization in unimodal and multimodal settings. We conduct a controlled and systematic experiment between LVLM and their LLMs counterparts to assess the effect of multimodal inputs on data memorization. Our analysis, uncovers that multimodal inputs exacerbate data memorization in LVLM, highlighting data memorization as a critical safety vulnerability, particularly for multimodal LVLM architectures.

success rate of ~60% (bottom-left subfigure). When the injection ratio exceeds 2%, the success rate becomes consistently high across models and tasks reaching over 90%. These results demonstrate that LVLM readily over-memorize injected concepts during fine-tuning, causing hallucinations. This vulnerability is particularly concerning, as it shows that standard LVLM fine-tuning can be destabilized with only a few injected samples. This finding comprehends our understanding of LVLM vulnerabilities by revealing a clearer root cause behind certain attacks such as ShadowCast [20], and it provides a valuable pathway toward an effective defenses. More broadly, the newly identified vulnerability exposes a novel risk, as adversaries may exploit it to develop new classes of attacks. We discuss these broader implications in detail in a later section.

> ***Finding 1:*** LVLMs tend to over-memorize fine-tuning concepts, leading to hallucinations in fine-tuned models.

3.3 Multimodal data exacerbate data memorization in LVLM

In Sec. 3.2, we identify data memorization as a previously unknown yet critical vulnerability of LVLM. In this section, we extend our investigation to further understand the data memorization in the multimodal setting of LVLMs when comparing with the unimodal setting. Through a controlled, systematic experiment between LVLM and their LLM-only counterparts, we show that multimodal inputs exacerbate data memorization effects in LVLM. This finding further underscores that data memorization is a critical safety vulnerability, particularly for multimodal LVLM architectures.

Experimental Design. To examine the impact of input modality on data memorization, we design a controlled and comparable experimental framework that contrasts LVLM (LLaVA 1.5 7B) with their unimodal, LLM-only counterparts (Vicuna 1.5 7B). Both models share the same underlying language backbone (Vicuna 1.5 7B), with the only architectural difference being the addition of a vision encoder in LVLM. This ensures that any observed differences in behavior can be attributed solely to the multimodal setting. From a dataset point of view, we follow our setup in Sec. 3.2 on the Biden-to-Trump task for the LVLM experimental setup. To ensure a fair and systematic comparison, we construct an equivalent setup for LLMs using only text. Particularly, in both setups, models are fine-tuned using LoRA on similarly sized datasets (3.5k samples): Sub-CC-Aligned [25] for LVLM (image-text) and Sub-Alpaca [16] for LLMs (text-only). In both settings, during the fine-tuning, we inject a small number of poisoning samples containing Biden content. For the LVLM setup, we use Biden image-text pairs provided in [20] but *without visual perturbations*. For the LLMs setup, we collect a comparable number of Biden text-only data points.

During evaluation, both models are presented with Trump-related queries: LVLM receive image-text pairs of Trump, while LLMs are given text-only questions about Trump. The aligned response is expected to mention Trump and avoid referencing Biden. In contrast, a hallucinated response incorrectly mentions Biden. By maintaining consistency across datasets, model sizes, fine-tuning methods, and poisoning content, our design ensures that the only variable under investigation is the data modality. This allows us to directly assess the extent to which multimodal inputs exacerbate memorization in LVLM.

Experimental results. As shown in Fig. 2-left, the attack accuracy of fine-tuned LVLM increases significantly once the injection ratio exceeds about 1%, rising from nearly 0% to over 90%. In contrast, the pre-trained LVLM are not hallucinated. This drastic jump indicates that multimodal LVLM can quickly memorize a number of injected images with the same destination concept during fine-tuning, allowing the injected content to strongly influence the fine-tuned LVLM' responses.

In comparison, the Fig. 2-right shows that when the same injection strategy is applied to unimodal LLMs (text-only), both pre-trained and fine-tuned models remain robust, with small attack success rates, even when the injected data ratio is up to 5%. Importantly, since both LVLM and LLMs share the same language backbone, model size, fine-tuning method, dataset size, and poisoning content, the observed differences can be directly attributed to the presence of a multimodal setting. The integration of visual information introduces additional pathways for memorization, making LVLM more susceptible to poisoning. We hypothesize that the added visual modality complicates the optimization landscape, increasing the risk of overfitting to spurious correlations and triggering memorization vulnerabilities.

> ***Finding 2:*** Multimodal data exacerbate data memorization in LVLM, highlighting data memorization as a critical safety vulnerability, particularly for multimodal LVLM architectures

4 Implications of Data Memorization for LVLM Vulnerability

In Sec. 3, we identify data memorization as a novel vulnerability in LVLM during fine-tuning. In this section, we discuss its implications for LVLM vulnerability. First, in Sec. 4.1, we show that data memorization is an overlooked yet critical factor underlying the SOTA visual perturbation-based data poisoning attacks on LVLM [20]. This explains why no effective defenses exist, as they fail to address the issue of data memorization. We then propose a simple yet effective defense, RejectShield, which significantly mitigates the attack. Second, in Sec. 4.2, we highlight that data memorization is a general and concerning vulnerability in LVLM. For example, an adversary could exploit this weakness to launch novel attack scenarios, which we propose a LLM-based monitoring defense. Together, RejectShield and LLM-based monitoring provide a comprehensive safeguard for LVLM fine-tuning datasets. Broadly, we highlight the long-term safety implications of our findings for LVLMs.

4.1 How Data Memorization Drives SOTA Data Poisoning Vulnerabilities

Inspired by our findings in Sec. 3, we show that data memorization is a key, overlooked factor behind the SOTA visual perturbation-based data poisoning attacks attack [20] explaining why no effective defenses exist, as they fail to address the issue of data memorization. To address this, we propose RejectShield, a rejection-based defense that effectively mitigates the attack.

4.1.1 Uncovering SOTA Data Poisoning Attack on LVLMs: Why Current Defenses Fail. The SOTA data poisoning attack, ShadowCast [20], justifies its success solely to the visual perturbation in poisoned examples. Consequently, existing defenses therefore apply SOTA purification methods from vision-only models to purify x_p. However, these purification-based defenses remain ineffective. In what follows, we explain this failure based on our findings on data memorization. Attack details can be found in the Sec. 3.1 and the ShadowCast paper [20].

Even without adversarial visual perturbations, the attack can still achieve high success rates through data memorization. The detailed results are shown in Fig. 1. As discussed in Sec. 3.2, attack success rate of our controlled experiment (red) variant closely approaches that of the standard "ShadowCast Attack", particularly when the poisoned ratio exceeds 2%. Notably, in the *Trump-to-Biden* task, the attack success rate of the controlled experiment variant nearly matches that of the standard "ShadowCast Attack". This suggests that, beyond the small poisoned-sample regime, adversarial visual perturbation contributes little to the attack's effectiveness. Instead, the model likely memorizes a number of these injected samples with the same destination concept during fine-tuning and produces the attacker's target response with very high success rates.

Data memorization explains why existing purification-based defenses are ineffective. Second, we find that purification-based defenses ("DiffPure" and "DiffJPEG") are only effective when the poisoning ratio is very low (≤1%). In this regime, these defenses can reduce the attack success rate to below 20%. However, once the poisoning ratio increases to 3% or higher, these defenses become ineffective, with attack success rates significantly increasing and becoming comparable to those of the standard "ShadowCast Attack". This is a surprising result, especially considering that the original justification for the success of ShadowCast [20] attributes its success solely to adversarial visual perturbations. Despite this assumption, purifying the visual perturbations is not an effective defense. Our new finding on the critical role of data memorization in the success of ShadowCast provides a clear explanation. Even when ideal purification is applied and adversarial perturbations are perfectly removed, the model can still memorize the poisoned captions during fine-tuning. As a result, it can still produce the attacker's target responses with very high success rates. Consequently, purification-based defenses are insufficient to mitigate the attack.

The transferability property of ShadowCast attack is strongly attributed to Data Memorization. ShadowCast attack is transferable [20], where poisoned samples crafted using one model can effectively poison other models. In their work, this property is originally attributed to adversarial transferability in vision models [9, 11]. However, our investigation of the role of data memorization in the ShadowCast attack provides a new perspective on this transferability. To gain deeper insight, we conduct a similar controlled experiment on ShadowCast's transferability by attacking MiniGPT4-v2 [25] using poisoned samples generated by LLaVA v1.5 7B (i.e., "Our controlled experiment"). We compare this with a baseline setup using unperturbed images (i.e., "ShadowCast Attack: LLaVA v1.5 -> MiniGPT4-v2"). The results are presented in Fig. 1-bottom. In contrast to the original justification [20], the results show that data memorization is the critical cause behind the success of the ShadowCast attack in this setup, while the transferability of visual perturbations plays a minimal role. The attack achieves high success rates on MiniGPT4-v2 even without visual perturbations, indicating that data memorization is the key. While the role of perturbation transferability is significant in very low poisoned ratios (≤1%), the difference diminishes as the poisoned ratio increases, with both scenarios reaching similarly high attack success rates. This observation is consistent with our findings and provides a clearer understanding of the success of previous results on black-box settings in [20].

> ***Finding 3:*** Data memorization during fine-tuning is the overlooked yet critical cause of ShadowCast's effectiveness. This oversight limits our understanding of the attack, resulting in ineffective defenses until now.

4.1.2 RejectShield: A Rejection-Based Defense Against ShadowCast Attack. In Sec. 4.1.1, we discover that data memorization is the previously unknown but critical cause of ShadowCast attack effectiveness. This suggests that even an ideal purifier that recovers the exact clean image x_d cannot effectively defend against ShadowCast attack. Inspired by our finding, we introduce *RejectShield*, a novel rejection-based defense to reject poisoned examples. Instead of attempting to purify or reconstruct each x_p as existing purification-based defenses, RejectShield employs an adversarial detector $f_{\text{adv}} : x \mapsto \{0, 1\}$, which detects whether an input image

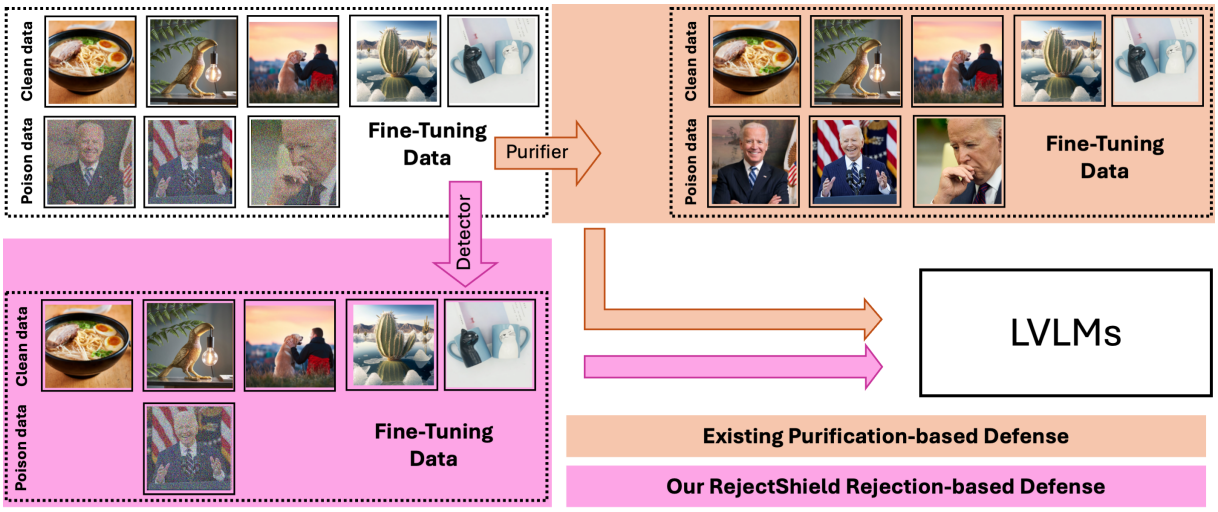

Figure 3: Our RejectShield defense vs. Existing defenses. While the SOTA LVLM data poisoning attack, attributes their success primarily to visual perturbations [20]. Consequently, existing defenses focus on purify such visual perturbations. However, our analysis reveals that data memorization is a previously unknown yet critical cause of ShadowCast's effectiveness. This insight explains why existing purification-based defenses that solely target visual perturbations are insufficient. Building on this novel finding, we propose RejectShield, the first rejection-based defense for LVLM data poisoning attacks that significantly mitigates the success of such attacks.

has likely been adversarially manipulated. *We emphasize that we follow the idea of [19], such that adversarial detection is generalizable to different types of adversarial perturbation. Importantly, no ShadowCast poison data is needed to train our detector.* We then filter the fine-tuning set as below and perform fine-tuning exclusively on $\mathcal{D}'_{\text{clean}}$.

$$\mathcal{D}'_{\text{clean}} = \{(x, y) \in \mathcal{D}_{\text{train}} : f_{\text{adv}}(x) = 0\},$$

RejectShield offers a simple yet effective defense against ShadowCast data poisoning in LVLM, even without using any ShadowCast poison data to train the detector.

Experimental Setup. We strictly follow the ShadowCast attack setups in [20] and use their open-source code for the implementations. Due to the space constrain, we present the main setups on two common LVLM, LLaVA 1.5 7B and MiniGPT4-2, with Label Attacks Category. For f_{adv} in our RejectShield defense framework, we adapt the adversarial detector from [19].

Experimental Results. The defense results are shown in Fig. 1 and Tab. 1. First, in Fig. 1, by directly targeting the root cause of data memorization through correctly rejecting poisoned samples, our RejectShield significantly outperforms existing defenses. Our defense can reduce attack success rates by up to 99%. Notably, under high poison ratios, where existing defenses fail and the attack success approaches the no defense baseline (i.e., ShadowCast Attack), RejectShield is still a strong defense. Second, RejectShield accurately accepts clean samples, resulting in model utilities comparable to the No Defense (i.e., ShadowCast Attack) and Clean Model (i.e., poison ratio = 0%) settings as shown in Tab. 1. This demonstrates that RejectShield effectively mitigates ShadowCast attacks with minimal sacrifice of model utilities.

4.2 Data Memorization is a general and concerning vulnerability in LVLM

Beyond poisoning, our findings highlight data memorization as a general, fundamental vulnerability of LVLMs, exposing new attack surfaces and pointing toward pathways for principled defenses.

The earlier stage of memorization-based attacks. Our investigation reveals that data memorization during LVLM fine-tuning exposes a general and critical vulnerability that adversaries can exploit. Such a memorization-based attack is particularly concerning, as adversaries need only standard fine-tuning procedures and the injection of benign destination data to trigger targeted model behaviors. Moreover, they target the crucial fine-tuning stage and do not rely on suspicious features in poisoned samples (such as visual perturbations used by ShadowCast). Without awareness of our discovered data memorization vulnerability, such attacks can stealthily cause fine-tuned LVLMs to hallucinate. The results of the memorization-based attack are illustrated as our controlled experiment in Fig. 1.

Despite their sophistication, memorization-based attacks typically require injecting (slightly) more samples than ShadowCast to achieve high success, though the absolute number remains small (i.e., less than ~ 5%). To counter such attacks, we propose *MemDefense*, an LLM-powered monitoring tool for safeguarding LVLM fine-tuning dataset curation. MemDefense uses a carefully designed prompt informed by our findings on data memorization, to detect overrepresented concepts. We evaluate MemDefense on the *Trump-to-Biden* task, analyzing fine-tuning text for memorization signals. The corresponding defense results are shown in Fig. 4. MemDefense flags "the current U.S. president Joe Biden" as highly overrepresented concepts likely to induce targeted hallucinations. These suspicious examples are then filtered from the fine-tuning data.

Table 1: Model utility comparison. Following ShadowCast setups [20], we report the model utility on VizWiz [4] and GQA [5] benchmarks. We compare ShadowCast Attack and our RejectShield defense. The results show that applying our RejectShield defense for LVLM fine-tuning primarily preserves the resulting model's utility.

Task	Defense	Benchmark	Poison Ratio (%)								
			0	0.1	0.3	0.6	0.9	1.4	2.9	4.3	5.7
Trump-to-Biden	No Defense	GQA	59.88	59.34	59.30	59.16	59.37	59.57	59.53	59.09	59.37
		VizWiz	56.42	56.15	56.30	56.31	56.56	56.22	56.31	55.98	56.43
	Ours	GQA	59.20	59.26	59.33	59.62	59.61	59.44	59.32	59.21	59.49
		VizWiz	55.78	55.85	56.07	56.02	55.99	55.77	55.83	56.15	55.82
Engine-to-Fuel	No Defense	GQA	59.88	59.22	59.37	59.29	59.29	59.50	59.74	59.39	59.59
		VizWiz	56.42	55.73	56.30	56.27	56.46	56.16	56.63	55.78	56.06
	Ours	GQA	59.26	59.21	59.26	59.12	59.32	59.19	59.15	59.13	59.17
		VizWiz	55.59	55.76	55.76	55.89	55.64	55.74	56.04	55.91	55.88

This highlights its practicality as a screening mechanism against memorization-based threats.

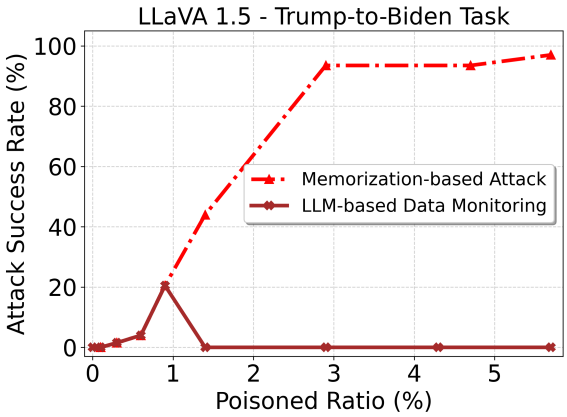

Figure 4: Defense results of leveraging an LLM as a monitoring tool to analyze the textual content of the fine-tuning dataset, guided by a prompt designed from our discovery on data memorization.

Towards Robust Defenses and Future Work. Building on our discovery of LVLMs' data memorization vulnerability, MemDefense offers an initial defense pathway and early discussion of memorization-based attacks. However, it may miss stealthier methods like ShadowCast, which achieve high success with fewer, more subtly injected samples, where our RejectShield remains crucial. Together, MemDefense and RejectShield provide a more comprehensive safeguard for LVLM fine-tuning datasets. As adversaries advance, poisoning attacks may become increasingly sophisticated, requiring fewer injected samples and potentially bypassing both language- and vision-based defenses. We therefore urge the community to develop stronger safeguards against these emerging threats.

5 Conclusion

This paper uncovers a critical vulnerability in LVLM: over-memorization during fine-tuning. We shows that LVLM tends to over-memorize fine-tuning concepts, leading to hallucinations in fine-tuned models. This effect is particularly pronounced in the multimodal setting of LVLM compared to unimodal models. Building on these findings, we uncover a comprehensive understanding of the SOTA LVLM data poisoning attack and explain why existing defenses fail. To mitigate this, we introduce RejectShield, a rejection-based defense that reduces attack success rates by up to 99% while nearly preserving model utility. More broadly, we discuss the implications of data memorization, highlighting potential new attack surfaces and pathways toward robust defenses.

Limitation. While our work sheds light on LVLM data poisoning and proposes effective defenses, it might not cover every scenario. Our experiments follow SOTA data poisoning LVLM setup [20] including common LVLMs (LLaVA-v1.5, MiniGPT4-v2), attack tasks, and clean datasets. Future work could explore this further setups.

Author contributions

Table 2: Tuyen (T) and Kecheng (K) contribute equally to this project. The detailed contributions are provided below.

	Idea	Code	Analysis	Writing
Finding 1	T	T	T	T, K
Finding 2	T	T	K	T, K
Finding 3	K	K	K	T, K

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Supplementary materials

The supplementary materials for this submission can be found in this link. The supplementary materials include:

- Additional results/writing (1.Supp.pdf)
- Pytorch-like code (2. Pytorch-like Code)
- LLM Analysis Data in Sec. 3.3 (3. LLM-Collected-Data)
- RejectShield Checkpoint (4. RejectShield-Checkpoint)

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